LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 11 Electricity

Chapter / Topic: Electricity · Week 6 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 25 minutes each: Day 1 on the day you study the chapter, then Day 2 and Day 3. Write answers in your notebook and check them with the key the same day. Copy every mistake into your error log.
2. Formulas: $Q = It$ · $V = W/Q$ · $V = IR$ · $R = \rho L/A$ · series $R = R_1 + R_2 + \ldots$ · parallel $1/R = 1/R_1 + 1/R_2 + \ldots$ · $P = VI = I^2R = V^2/R$ · $H = I^2Rt$ · $1\ \text{kWh} = 3.6 \times 10^6$ J

DAY 1: CURRENT, POTENTIAL DIFFERENCE AND OHM'S LAW

Q1. One ampere is equal to:

(a) 1 coulomb per second (b) 1 joule per second
(c) 1 joule per coulomb (d) 1 ohm per volt

Q2. At a constant temperature, the resistance of a wire does NOT depend on:

(a) its length
(b) the potential difference applied across it
(c) its area of cross-section
(d) the material it is made of

Q3. A current of 2 A flows through a wire for 3 minutes. The charge that flows is:

(a) 6 C (b) 90 C
(c) 360 C (d) 600 C

Q4. The potential difference between two points is the:

(a) charge flowing per unit time
(b) work done to move a unit charge from one point to the other
(c) current per unit resistance
(d) energy used per unit time

Q5. Draw the circuit symbols for: a cell, a battery, a closed plug key, a resistor, a rheostat (variable resistor), an ammeter and a voltmeter.

Q6. A potential difference of 6 V is applied across a 24 Ω resistor. Find the current. What will the current be if the potential difference is doubled?

Q7. Wire P has length L and area of cross-section A. Wire Q, made of the same material, has length 2L and area of cross-section 4A. Compare their resistances.

Q8. A nichrome wire 2 m long, with an area of cross-section of 0.5 mm^2, has a resistance of 4 Ω. Find the resistivity of nichrome.

DAY 2: RESISTORS IN SERIES AND PARALLEL

Q9. Resistors of 3 Ω, 6 Ω and 9 Ω are connected in series. The equivalent resistance is:

(a) 1.6 Ω (b) 6 Ω
(c) 18 Ω (d) 27 Ω

Q10. Resistors of 6 Ω and 3 Ω are connected in parallel. The equivalent resistance is:

(a) 2 Ω (b) 4.5 Ω
(c) 9 Ω (d) 18 Ω

Q11. In a parallel combination of resistors, which quantity is the same for every resistor?

(a) the current (b) the potential difference
(c) the resistance (d) the power

Q12. You have three 6 Ω resistors. How will you connect them to get an equivalent resistance of (a) 9 Ω, (b) 4 Ω?

Q13. Resistors of 5 Ω, 10 Ω and 15 Ω are connected in series to a 6 V battery. Find the current and the potential difference across each resistor.

Q14. Resistors of 10 Ω and 40 Ω are connected in parallel, and this combination is connected in series with a 2 Ω resistor and a 10 V battery. Find (i) the total resistance, (ii) the current from the battery, (iii) the current through the 10 Ω and 40 Ω resistors.

Q15. Which has the higher resistance: a 100 W bulb or a 60 W bulb, both rated for 220 V? Show by calculation.

DAY 3: HEATING EFFECT AND ELECTRIC POWER

Q16. **Assertion (A):** Electric power is measured in kilowatt-hours.
Reason (R): 1 kWh = 3.6×10^6 J.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q17. The heat produced in a resistor is $H = I^2Rt$. If the current is doubled, the heat produced in the same time becomes:

(a) half (b) the same
(c) double (d) four times

Q18. In a circuit, a fuse is connected:

(a) in series with the device
(b) in parallel with the device
(c) between the earth wire and the neutral wire
(d) only to the earth wire

Q19. An electric kettle is rated 880 W, 220 V.

(i) Find the current it draws.

(ii) Find the resistance of its heating element.

(iii) Fuses of 2 A, 5 A and 15 A are available. Which one should be used for the kettle? Why?

Q20. (a) Starting from $P = VI$, show that $P = I^2R$ and $P = V^2/R$.
(b) A motor draws a current of 3 A from a 220 V supply. Find its power, the energy it uses in 5 hours (in kWh), and the cost if electricity costs ₹6 per unit.

Q21. The 100 W and 60 W bulbs from Day 2 are connected in series to a 220 V supply. Which one glows brighter? Explain.

~ Write the formula, substitute, then the unit. That's 3 marks every time! ~